

Aditi Shukla

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EDUCATION

United College of Engineering and Research

2024 – 2028

Bachelor of Technology (B.Tech) in Computer Science & Engineering

First Year SGPA: 8.29

TECHNICAL SKILLS

Languages: Python, Java, C

AI/ML: Machine Learning fundamentals, Generative AI, Computer Vision, YOLO, MediaPipe

Robotics/Embedded: ROS 2, Raspberry Pi, Arduino, ESP32/NodeMCU, sensor integration, embedded programming

Web Development: React.js, Next.js, Node.js/Express, Tailwind CSS, Firebase

APIs & Tools: Gemini API, Whisper/AssemblyAI, , Git/GitHub, Linux

PROJECTS

RakshaSetu AI Python, ROS 2, OpenCV, MediaPipe, YOLO, Raspberry Pi, Arduino

- Engineered an AI-powered elder-care companion robot that combines computer vision, voice interaction, and embedded control for emotional support and safety monitoring
- Designed a multi-layer hardware architecture (chassis, power deck, compute deck, sensor/head unit) integrating Raspberry Pi 4/5, Arduino Nano, NodeMCU/ESP32, and an L298N motor driver
- Built hand-gestation recognition and a fall/pain-detection concept using YOLO and MediaPipe, tuned for constrained onboard compute
- Implemented ROS 2-based motor control and ultrasonic obstacle sensing, structuring perception, control, and communication as modular nodes

SignBridge AI (Sanket Setu AI) React.js, TypeScript, Node.js, Gemini API, MediaPipe, Firebase

- Developed a full-stack accessibility platform serving deaf/mute, visually impaired, and dyslexic users through a single unified interface, as part of a 4-member team, **HER TECH HIVE**
- Built real-time sign language recognition using MediaPipe hand-tracking integrated with a Gemini API translation pipeline
- Delivered voice-based navigation and a dyslexia-friendly reading mode, built and shipped under hackathon time constraints
- Selected as a finalist among the **Top 66 of 260+ teams** at UHACK 4.0; awarded the **Women Tech Empowerment Award (Rs. 5,000)** for project impact and execution

Bomb Quiz Game Java (Swing/AWT)

- Built a gamified Java desktop application using multithreading to drive a real-time countdown mechanic under time pressure
- Applied inheritance and encapsulation to support dynamic question categories and evolving game states
- Achieved zero runtime crashes across testing through robust exception handling and disciplined memory management

TRAINING & COMPETITIONS

IIIT Allahabad – AIR.2026 Offline Training

June 22 – July 19, 2026

- Completed an intensive offline training program in Artificial Intelligence, Machine Learning, Robotics, Embedded Systems, Python, ROS 2 basics, and simulation environments

AIRena 2026 Robotics Challenge

- Applied robotics navigation concepts including line following, maze navigation, ultrasonic obstacle detection, PID control, and micro-ROS architecture understanding

LEADERSHIP & EXTRA-CURRICULARS

- **Team Lead, Her Tech Hive:** Leading a cross-functional team on RakshaSetu AI , coordinating hardware, software, and presentation workstreams while liaising with a faculty mentor
- **Leadership:** Former Head Girl of school, managing student councils and organizing large-scale campus events, Current Class Representative
- **Communication:** Experience in anchoring and event coordination for college occasions; volunteered for **Abhivyakti** and multiple college events
- **Arts:** Active member of singing club **Pravah**